



MS/MA IN ECONOMICS

DYNAMIC STOCHASTIC MODELS

Lecture 03

Variational Methods

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1 The problem

We look for the stationary (steady-state) equilibrium of a competitive economy in which the representative household has preferences

$$u(c_t^i, 1 - h_t^i) = \log c_t^i + A \log(1 - h_t^i),$$

and production is Cobb–Douglas,

$$f(K, H) = \gamma K^\theta H^{1-\theta},$$

with parameters A, γ, θ, β and δ . The goal is to obtain the steady-state values of output $\bar{Y}$, consumption $\bar{C}$, capital $\bar{K}$ and labor $\bar{H}$.

2 The steady state in its generic form

2.1 Equilibrium conditions for the individual level

The problem fits in the general form seen in class for a competitive economy that includes the optimal leisure-consumption decision made by the representative household (unlike de Robinson Crusoe's case). Thus, we take the equilibrium conditions obtained from the first-order conditions:

$$\frac{u_h(c_t^i, 1 - h_t^i)}{u_c(c_t^i, 1 - h_t^i)} = f_h(K_t, H_t) \quad (\text{labor-leisure})$$

$$\frac{u_c(c_t^i, 1 - h_t^i)}{u_c(c_{t+1}^i, 1 - h_{t+1}^i)} = \beta [f_k(K_{t+1}, H_{t+1}) + (1 - \delta)] \quad (\text{Euler for capital})$$

$$k_{t+1}^i = (1 - \delta)k_t^i + f_h(K_t, H_t)h_t^i + f_k(K_t, H_t)k_t^i - c_t^i \quad (\text{Budget constraint})$$

2.2 Aggregate equilibrium conditions

The aggregation conditions are:

$$K_t = \int_0^1 k_t^i di \quad H_t = \int_0^1 h_t^i di \quad C_t = \int_0^1 c_t^i di$$

The unit mass of agents, continuously indexed from zero to one, implies that, given we are operating with a representative agent

$$c_t^i = C_t \quad y \quad h_t^i = H_t \quad \text{para todo } i.$$

Additionally, we know from the neoclassical framework that production has constant returns to scale (production function is first-degree homogeneous). Therefore, we can use Euler's theorem for homogeneous functions

$$f_h(K_t, H_t) H_t + f_k(K_t, H_t) K_t = f(K_t, H_t).$$

Replacing these results in the equilibrium conditions yields the aggregate equilibrium conditions:

$$\begin{aligned}\frac{u_h(C_t, 1 - H_t)}{u_c(C_t, 1 - H_t)} &= f_h(K_t, H_t) \\ \frac{u_c(C_t, 1 - H_t)}{u_c(C_{t+1}, 1 - H_{t+1})} &= \beta [f_k(K_{t+1}, H_{t+1}) + (1 - \delta)] \\ K_{t+1} &= (1 - \delta)K_t + f(K_t, H_t) - C_t\end{aligned}$$

2.3 Steady state

Imposing that all aggregate variables are constant ($C_t = \bar{C}$, $H_t = \bar{H}$, $K_t = \bar{K}$), we have three equations in the three unknowns $\bar{C}$, $\bar{H}$, $\bar{K}$:

$$\frac{u_h(\bar{C}, 1 - \bar{H})}{u_c(\bar{C}, 1 - \bar{H})} = f_h(\bar{K}, \bar{H}) \quad (\text{labor-leisure}) \quad (1)$$

$$\frac{1}{\beta} - (1 - \delta) = f_k(\bar{K}, \bar{H}) \quad (\text{capital}) \quad (2)$$

$$\bar{C} = f(\bar{K}, \bar{H}) - \delta \bar{K} \quad (\text{resource constraint}) \quad (3)$$

The first equation is intratemporal (it equates the marginal rate of substitution between leisure and consumption to the real wage); the second is the intertemporal (Euler) condition evaluated at a constant capital stock; the third is the economy-wide resource constraint, which in steady state says that output net of depreciation replacement equals consumption.

3 Application to the particular case

3.1 Marginal utilities and marginal products

Marginal utilities. With $u(c, 1-h) = \log c + A \log(1-h)$, differentiate with respect to each argument:

$$u_c = \frac{\partial u}{\partial c} = \frac{1}{c}, \quad u_h = \frac{\partial u}{\partial h} = -\frac{A}{1-h}.$$

The negative sign in u_h arises because a higher h lowers leisure $1-h$, and utility is increasing in leisure. Forming the ratio that appears on the left-hand side of (1) (the marginal rate of substitution, taken in absolute value):

$$\frac{u_h}{u_c} = \frac{A/(1-h)}{1/c} = \frac{A c}{1-h}.$$

Marginal products. With $f(K, H) = \gamma K^\theta H^{1-\theta}$, it is convenient to work with the *capital intensity*

$$k \equiv \frac{K}{H}.$$

Because f is homogeneous of degree one, its partial derivatives depend only on k . Differentiating,

$$\begin{aligned} f_k &= \frac{\partial f}{\partial K} = \theta \gamma K^{\theta-1} H^{1-\theta} = \theta \gamma \left(\frac{K}{H} \right)^{\theta-1} = \theta \gamma k^{\theta-1}, \\ f_h &= \frac{\partial f}{\partial H} = (1 - \theta) \gamma K^\theta H^{-\theta} = (1 - \theta) \gamma \left(\frac{K}{H} \right)^\theta = (1 - \theta) \gamma k^\theta. \end{aligned}$$

Likewise, output per worker depends only on k :

$$\frac{Y}{H} = \frac{f(K, H)}{H} = \gamma \frac{K^\theta H^{1-\theta}}{H} = \gamma \left(\frac{K}{H} \right)^\theta = \gamma k^\theta.$$

3.2 The capital-labor ratio $k = \bar{K}/\bar{H}$

The key observation is that the capital equation (2) involves neither consumption nor labor separately, only the ratio k . Substituting f_k ,

$$\theta \gamma k^{\theta-1} = \frac{1}{\beta} - 1 + \delta.$$

Solving for k ,

$$k^{\theta-1} = \frac{\frac{1}{\beta} - 1 + \delta}{\theta \gamma} \quad \Longrightarrow \quad \boxed{k = \left(\frac{\frac{1}{\beta} - 1 + \delta}{\theta \gamma} \right)^{\frac{1}{\theta-1}}}$$

This is the backbone of the whole solution: the long-run capital intensity is **determined on its own**, independently of the labor and consumption choices.

Once k is known, all quantities measured per unit of labor are fixed. Output per worker follows directly:

$$\frac{\bar{Y}}{\bar{H}} = \gamma k^\theta.$$

Consumption per worker comes from the resource constraint (3). Divide it by $\bar{H}$ and use $\bar{K}/\bar{H} = k$:

$$\frac{\bar{C}}{\bar{H}} = \frac{f(\bar{K}, \bar{H})}{\bar{H}} - \delta \frac{\bar{K}}{\bar{H}} = \frac{\bar{Y}}{\bar{H}} - \delta k = \gamma k^\theta - \delta k.$$

The steady-state wage (needed next) is also a function of k only:

$$\bar{w} = f_h(\bar{K}, \bar{H}) = (1 - \theta) \gamma k^\theta.$$

3.3 The labor–leisure condition pins down $\bar{H}$

Substitute the marginal rate of substitution from Step 1 into the labor–leisure condition (1):

$$\frac{A\bar{C}}{1-\bar{H}} = f_h(\bar{K}, \bar{H}).$$

To isolate $\bar{H}$, write consumption as its per-worker value times the level of labor, $\bar{C} = \left(\frac{\bar{C}}{\bar{H}}\right) \bar{H}$:

$$\frac{A\left(\frac{\bar{C}}{\bar{H}}\right)\bar{H}}{1-\bar{H}} = f_h.$$

Both $\frac{\bar{C}}{\bar{H}}$ and f_h are known constants (functions of k only), so this is a single linear equation in $\bar{H}$. Multiply through by $(1-\bar{H})$:

$$A\left(\frac{\bar{C}}{\bar{H}}\right)\bar{H} = f_h(1-\bar{H}) = f_h - f_h\bar{H}.$$

Collect the $\bar{H}$ terms on one side:

$$\bar{H}\left[A\left(\frac{\bar{C}}{\bar{H}}\right) + f_h\right] = f_h,$$

and solve:

$$\bar{H} = \frac{f_h}{A\left(\frac{\bar{C}}{\bar{H}}\right) + f_h} = \frac{(1-\theta)\gamma k^\theta}{A(\gamma k^\theta - \delta k) + (1-\theta)\gamma k^\theta}.$$

Since $0 < \bar{H} < 1$ whenever the per-worker consumption is positive, this yields an admissible labor supply.

3.4 Recovering the levels

With $\bar{H}$ and the per-worker ratios in hand, every level scales linearly:

$$\begin{aligned}\bar{K} &= k \bar{H}, \\ \bar{Y} &= \left(\frac{\bar{Y}}{\bar{H}}\right) \bar{H} = \gamma k^\theta \bar{H}, \\ \bar{C} &= \left(\frac{\bar{C}}{\bar{H}}\right) \bar{H} = (\gamma k^\theta - \delta k) \bar{H}, \\ \bar{I} &= \delta \bar{K} = \delta k \bar{H}.\end{aligned}$$

As a consistency check, the resource constraint must hold at the level of aggregates:

$$\bar{C} + \bar{I} = (\gamma k^\theta - \delta k) \bar{H} + \delta k \bar{H} = \gamma k^\theta \bar{H} = \bar{Y}. \quad \checkmark$$

4 Numerical solution

We now evaluate the closed-form expressions from the previous section at the parameter values given in the assignment. We then compare the results with those obtained using Matlab.

$$A = 1.72, \quad \gamma = 3, \quad \theta = 0.56, \quad \beta = 0.98, \quad \delta = 0.06.$$

Capital intensity. From the capital condition,

$$\begin{aligned} \frac{1}{\beta} - 1 + \delta &= \frac{1}{0.98} - 1 + 0.06 = 0.0804082, \\ \kappa = \frac{\bar{K}}{\bar{H}} &= \left(\frac{0.0804082}{\theta \gamma} \right)^{\frac{1}{\theta-1}} = \left(\frac{0.0804082}{0.56 \cdot 3} \right)^{\frac{1}{0.56-1}} = 1000.0478. \end{aligned}$$

Per-worker quantities.

$$\begin{aligned} \frac{\bar{Y}}{\bar{H}} &= \gamma \kappa^\theta = 3 \cdot 1000.0478^{0.56} = 143.5929, \\ \frac{\bar{C}}{\bar{H}} &= \gamma \kappa^\theta - \delta \kappa = 143.5929 - 0.06 \cdot 1000.0478 = 83.5900, \\ \bar{w} = f_h &= (1 - \theta) \gamma \kappa^\theta = 0.44 \cdot 143.5929 = 63.1809. \end{aligned}$$

Labor.

$$\bar{H} = \frac{f_h}{A \left(\frac{\bar{C}}{\bar{H}} \right) + f_h} = \frac{63.1809}{1.72 \cdot 83.5900 + 63.1809} = 0.3053.$$

Levels.

$$\begin{aligned} \bar{K} &= \kappa \bar{H} = 1000.0478 \cdot 0.3053 = 305.3015, \\ \bar{Y} &= \frac{\bar{Y}}{\bar{H}} \bar{H} = 143.5929 \cdot 0.3053 = 43.8370, \\ \bar{C} &= \frac{\bar{C}}{\bar{H}} \bar{H} = 83.5900 \cdot 0.3053 = 25.5189, \\ \bar{I} &= \delta \bar{K} = 0.06 \cdot 305.3015 = 18.3181. \end{aligned}$$

The resource constraint is satisfied: $\bar{C} + \bar{I} = 25.5189 + 18.3181 = 43.8370 = \bar{Y}$.

5 Dynare code

The steady state is computed in Dynare through a `steady_state_model` block, which evaluates the closed-form expressions derived above directly, without numerical iteration. The full model is written out in the `model` block so that Dynare can independently verify, via the `steady` command, that the reported values satisfy every equilibrium equation.

```
1 // Variables
2 var C H K Y I; // consumption, labor, capital, output, investment
3
4 // Parameters
5 parameters A gamma theta beta delta;
6 A = 1.72;
7 gamma = 3;
8 theta = 0.56;
9 beta = 0.98;
10 delta = 0.06;
11
12 // Model
13 model;
14 // Production
15 Y = gamma * K(-1)^theta * H^(1-theta);
16
17 // Labor-leisure (intratemporal): u_h/u_c = f_h
18 A*C/(1-H) = (1-theta)*Y/H;
19
20 // Euler equation (intertemporal)
21 (1/C) = beta*(1/C(+1))* ( theta*Y(+1)/K + (1-delta) );
22
23 // Capital accumulation
24 K = (1-delta)*K(-1) + I;
25
26 // Resource constraint
27 Y = C + I;
28 end;
29
30 // Steady state: closed form (Step 1)
31 steady_state_model;
32 kappa = ( (1/beta - 1 + delta) / (theta*gamma) )^(1/(theta-1));
33 H = (1-theta)*gamma*kappa^theta
34 / ( A*(gamma*kappa^theta - delta*kappa) + (1-theta)*gamma*kappa^theta );
35 K = kappa*H;
36 Y = gamma*kappa^theta*H;
37 C = (gamma*kappa^theta - delta*kappa)*H;
38 I = delta*K;
39 end;
40
```

```

41 // Compute and report the steady state
42 steady;

```

STEADY-STATE RESULTS:

C	25.5189
H	0.305287
K	305.302
Y	43.837
I	18.3181

Cross-check: analytical vs. Dynare

Both routes—hand evaluation of the closed-form formulas and Dynare’s `steady_state_model`—produce identical values, confirming that the algebra and the code are mutually consistent:

Variable	Symbol	Value
Output	$\bar{Y}$	43.8370
Consumption	$\bar{C}$	25.5189
Capital	$\bar{K}$	305.3015
Labor	$\bar{H}$	0.3053
Investment	$\bar{I}$	18.3181

The requested steady-state values are therefore

$$\bar{Y} \approx 43.84, \quad \bar{C} \approx 25.52, \quad \bar{K} \approx 305.30, \quad \bar{H} \approx 0.31.$$

Remark. The equilibrium labor supply $\bar{H} \approx 0.31$ implies the household devotes roughly one third of its available time to work. The capital intensity $\kappa = \bar{K}/\bar{H} \approx 1000$ is large because of the chosen parameters (in particular the productivity level $\gamma = 3$ combined with the capital share $\theta = 0.56$), which pushes the capital stock into the hundreds even though labor is a fraction of one.